Shourya Pandey

Curriculum Vitae

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Research Interests

- Computational Complexity
- Combinatorics and Graph Theory
- Automata Theory
- Algorithm Design

Education

August, 2020 **B.Tech with Honors, Computer Science and Engineering** *GPA: **9.10/10***
Indian Institute of Technology Bombay, Maharashtra, India
Minor in *Mathematics*

Publications

- Prasad Chaugule, Nutan Limaye, and *Shourya Pandey* "**Variants of the Determinant Polynomial and VP-completeness**," *under review in Symposium on Theoretical Aspects of Computer Science (STACS) 2021* [*link*]
- Prasad Chaugule, Nutan Limaye, and *Shourya Pandey*, "**Monotone Arithmetic Classes in the Border World**," *draft in progress*

Research Experience

Spring 2020 **Natural VP-complete polynomial families** [*paper*] *Research Project*
Prof. Nutan Limaye IIT Bombay

- Explored candidate **VP-complete polynomial families** by considering variants of known VBP-complete or VNP-complete polynomial families, such as the coloured Tutte polynomial on special graph families
- Studied **Branching Programs** with **memory**, such as Stack Branching Programs (SBPs) and Read Once Branching Programs (RABPs), which model the classes VP and VNP respectively
- Worked on a natural variant of the determinant polynomial family complete for VP and **extended the variant** to another family of polynomials complete for VNP, where the completeness is under p-projections

Fall 2020 (In progress) **Monotone Arithmetic Classes and Border Complexity** *Research Project*
Prof. Nutan Limaye IIT Bombay

- Working on the computational power of **monotone** and **border monotone** classes in the arithmetic world
- Extending the result in another paper that equates the powers of monotone VBP and border monotone VBP classes to other algebraic classes such as VF, VP, and VNP

Summer 2019 **Portfolio Optimization and Hierarchical Risk Parity** *Research Internship*
Amitabh Iyer, Abhijit Lal WorldQuant Research, India

- Implemented various portfolio optimization Models such as the Markowitz model and the **Hierarchical Risk Parity (HRP)** model; tested the performances of these models using real and random data
- Improved the performance of the existing implementation of portfolio optimization by **over 100 times**
- Built an **equity risk factor model** to attribute risk exposures taken on by equity investment strategies, and also developed a fast and interactive user-interface to analyze trading strategies across various categories

Summer 2018 **Strong Collapse for Persistence** *Research Project*
Prof. Jean-Daniel Boissonnat INRIA Sophia-Antipolis, France

- Worked on adding functions on the **GUDHI** (Geometric Understanding in Higher Dimensions) library, an open-source C++ library for Topological Data Analysis (TDA) and computational topology
- Analyzed **strong collapse**, which is a series of elementary collapse operations on sequences of simplicial complexes and which preserves the strong homotopy type of the simplicial complex
- Implemented a **fast and memory-efficient** code to compute the strong collapse for simplicial complexes in order to compute the persistent homology, which is **NP-hard** when the simplices are represented succintly

Key Scholastic Achievements

2020 Awarded the **Undergraduate Research Award** for outstanding research work in my senior year

2017 Received **Advanced Performer**'s grade (about top 1% of class) in Linear Algebra

Teaching Experience

Fall 2019 **Teaching Assistant** *CS207: Discrete Structures*

Conducted weekly tutorial sessions for a batch of more than 100 sophomores. Evaluated the answer scripts of the students and discussed the marking scheme of the examinations with the instructor.

Spring 2019 **Lab Assistant** *CS101: Computer Programming and Utilization*

Assisted in managing biweekly graded and ungraded lab sessions for a batch of more than 500 freshmen. Set biweekly coding assignments for the batch and invigilated lab tests and written tests.

Spring 2018 **Teaching Assistant** *MA106: Linear Algbera*

Offered help sessions for students who needed more help in the course. Evaluated answer scripts and helped the instructor design the grading policy.

Summer 2017 **Summer Teaching Assistant** *MA105: Calculus*

Kept classes and tutorials for a batch of around 30 students from IIT Bombay, IIT Goa, and IIT Dharwad for the summer offering of calculus for students who were required to improve their grades.

Olympiad Experience

Summer 2015 **Bronze medalist** [*profile*] *International Mathematical Olympiad (IMO)*

Stood **second at the IMO Training Camp**, India, in 2015. Represented India at the IMO 2015 at Chiang Mai and bagged a bronze medal. Was also provided a **one-time prize** from the Homi Bhabha Centre for Science Education (HBCSE), a national centre for the Tata Institute of Fundamental Research (TIFR).

Summer 2019 **Grader** *Various mathematical olympiads*

Volunteered to grade the scripts of over a 1000 students at the Indian National Mathematical Olympiad 2019 (**INMO**, which is analogous to the USAMO), around 60 students at the IMO Training Camp India 2019 (**IMOTC**, analogous to the USTST), and around 80 students who represented Indian for the Asian Pacific Mathematical Olympiad 2020 (**APMO**, which is an olympiad conducted across 40 countries).

Winter 2019 **Invitational Lecturer** *INMO Training Camp*

Taught the subjects of matching theory, enumerative combinatorics, and number theory at the **INMO Training Camp**, which is a camp for students in the Mumbai region who contest for the INMO.

Summer 2020 **Outreach in the pandemic**

Took the initiative to teach olympiad mathematics and theoretical computer science online to high school students. Conducted mock tests for the students to teach them new methods and test their proof-writing skills and mathematical maturity. Prepared handouts on certain topics in the fields of combinatorics and number theory, for the olympiad mathematics community to benefit from it.

Co-scholastics

Summer 2018, 2019, and 2020 **Mentor** *Summer of Science*

Worked alongside the Maths and Physics Club of IIT Bombay and **mentored around 5 freshmen** to read and develop interest in various topics such as data structures, algorithm design, graph theory, analytic number theory, coding theory, and competitive programming.

Fall 2019 **Invitational Speaker** *Maths and Physics Club, IIT bombay*

Presented a talk on complexity theory, and in particular introduced basic classes such as P, NP, PSPACE and EXP, introduced the notion of NP-hardness and NP-completeness, and gave some insights to the P vs NP problem, and discussed the possible implications and non-implications of P=NP in the modern world.

Spring 2020 **Sports Color** *IIT Bombay*

Received the Color, which is awarded to students for **exceptional contribution** to the institute sports throughout college; this was awarded to me because of **2 gold medals and 1 bronze medal** at the Inter-IIT **chess** tournaments, along with a **best player** of the tournament award in Inter-IIT 2018.

Fall 2017 and Spring 2018 **Sports Convener** *The Dark Knight Chess Club, IIT Bombay*

Organized chess camps and tournaments for the students and professors of the college. Helped organized a tournament between various colleges within the city in **Aavhan**, the biggest sports festival of IIT Bombay.

Select Course Projects

Spring 2019 **Approximation Algorithms for Metric TSP**
Prof. Rohit Gurjar — Perfect Matchings: Algorithms and Complexity
Analyzed various NP-complete problems and studied approximation algorithms for them. Identified and proved that the Travelling Salesman Problem (TSP problem) is **APX-hard**, and focused on special cases of the TSP problem, and in particular Metric TSP. Proved that **Metric TSP** is NP-complete, and provided the naive 2-approximation algorithm and the improved $3/2$-approximation algorithm (**Christofides' algorithm**) for the same. Proved independently that $3/2$ is the best constant provided by the Christofides' algorithm.

Spring 2020 **Tutte Polynomials; Monotone Arithmetic Classes**
Prof. Nutan Limaye — Research Project
Studied colored Tutte polynomials, a generalization of the **Tutte's polynomial** for Valiant's setting. Worked out graph families that are VNP-complete under p-projections. Computed the generalized Tutte polynomial for various sparse graphs such as the ladder graph, the wheel graph, and the fan graph, classified them in the classes of VF, VBP, VP or VNP, and tried to prove completeness of these polynomial families. Also worked on **border complexity class** and its relation to **monotone circuits**, and attempted to extend a paper which showed the equivalence of monotone VBP and border monotone VBP complexity classes.

Fall 2018 **Communication Complexity and The Indexing Problem**
Prof. Nutan Limaye — Introduction to Computational Complexity
Presented the idea of communication complexity between two parties and multiple parties and discussed the representations of balancing protocol and monochromatic rectangles. Also discussed about the techniques of rectangle covers, fooling sets, and ranks to get lower bounds on various problems. Talked about **randomized protocols** (both public and private coins), and used inequalities from **information theory** to get $\Omega(n)$ lower bounds on problems such as **pointer chasing** and with more focus on the **indexing problem**. Finally, mentioned an application of communication complexity to the domain of algebraic circuit complexity.

Spring 2017 **A Chess Engine**
Prof. Amitabh Sanyal — Abstractions and Paradigms of Computer Programming
Devised a chess engine on the functional programming language Racket. Used **minimax** algorithm along with **alpha-beta pruning** to decrease the number of nodes evaluated by the minimax algorithm in its search tree.

Relevant Coursework

Computer Science: Introduction to Complexity Theory, Topics in Computational Complexity, Perfect Matchings: Algorithms and Complexity, Graph Theory, Cryptography and Network Security, Formal Methods for Concurrent and Asynchronous Systems, Formal Verification and Specification of Programs, Automata Theory, Discrete Structures, Algorithm Design, Data Structures and Algorithms

Mathematics: Linear Algebra, Advanced Enumerative Combinatorics, Graph Theory, Introductory Number Theory, Basic Algebra, Multivariate Calculus, Real Analysis, Complex Analysis, Differential Equations, Numerical Analysis

References

- **Prof. Nutan Limaye** ✉
  Associate Professor, IIT Bombay
- **Prof Rohit Gurjar** ✉
  Associate Professor, IIT Bombay
- **Abhijit Lal** ✉
  Vice President, WorldQuant Research India